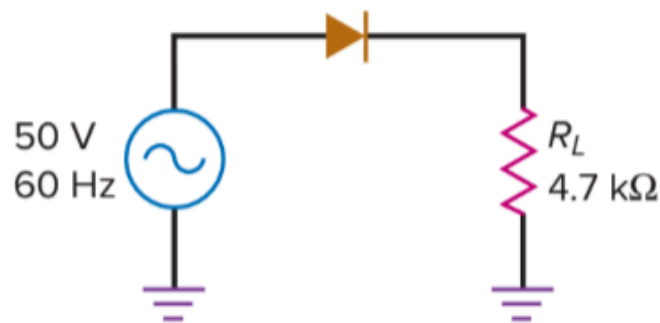
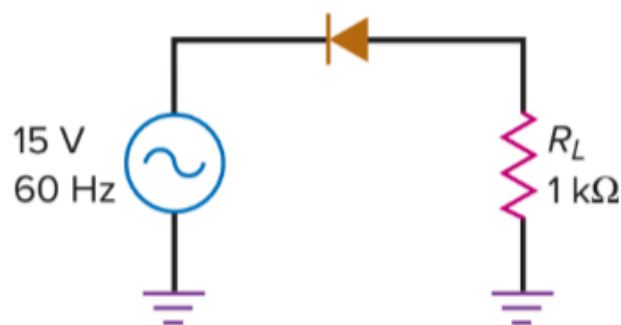


- 4-1** **MultiSim** What is the peak output voltage in Fig. 4-36a if the diode is ideal? The average value? The dc value? Sketch the output waveform.

Figure 4-36



(a)

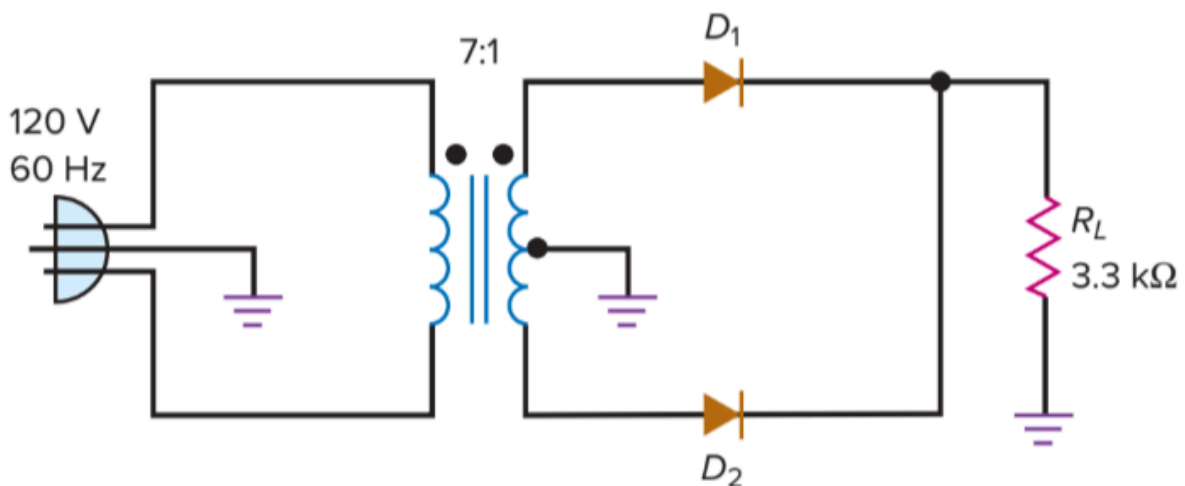


(b)

- 4-2** **MultiSim** Repeat the preceding problem for Fig. 4-36b.
- 4-3** **MultiSim** What is the peak output voltage in Fig. 4-36a using the second approximation of a diode? The average value? The dc value? Sketch the output waveform.
- 4-4** **MultiSim** Repeat the preceding problem for Fig. 4-36b.

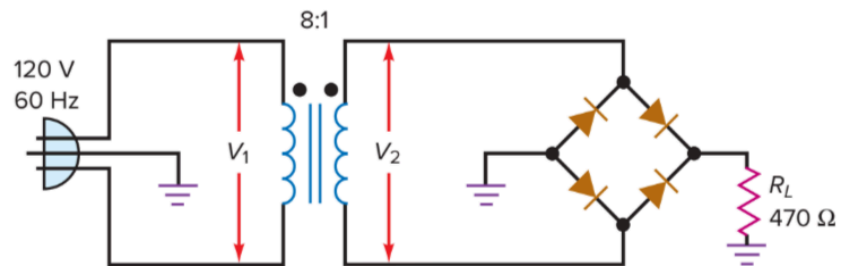
- 4-10** **MultiSim** What is the peak output voltage in Fig. 4-38 if the diodes are ideal? The average value? The dc value? Sketch the output waveform.
- 4-11** **MultiSim** Repeat the preceding problem using the second approximation.

Figure 4-38



- 4-12** **MultiSim** In Fig. 4-39, what is the peak output voltage if the diodes are ideal? The average value? The dc value? Sketch the output waveform.
- 4-13** **MultiSim** Repeat the preceding problem using the second approximation.
- 4-14** If the line voltage in Fig. 4-39 varies from 105 to 125 V_{rms} , what is the minimum dc output voltage? The maximum?

Figure 4-39



- 4-15** A half-wave signal with a peak of 20 V is the input to a choke-input filter. If $X_L = 1\text{ k}\Omega$ and $X_C = 25\ \Omega$, what is the approximate peak-to-peak ripple across the capacitor?
- 4-16** A full-wave signal with a peak of 14 V is the input to a choke-input filter. If $X_L = 2\text{ k}\Omega$ and $X_C = 50\ \Omega$, what is the approximate peak-to-peak ripple across the capacitor?
- 4-17** What is the dc output voltage and ripple in Fig. 4-40a? Sketch the output waveform.
- 4-18** In Fig. 4-40b, calculate the dc output voltage and ripple.
- 4-19** What happens to the ripple in Fig. 4-40a if the capacitance value is reduced to half?
- 4-20** In Fig. 4-40a, what happens to the ripple if the resistance is reduced to $500\ \Omega$?
- 4-21** What is the dc output voltage in Fig. 4-41? The ripple? Sketch the output waveform.
- 4-22** If the line voltage decreases to 105 V in Fig. 4-41, what is the dc output voltage?

Figure 4-40

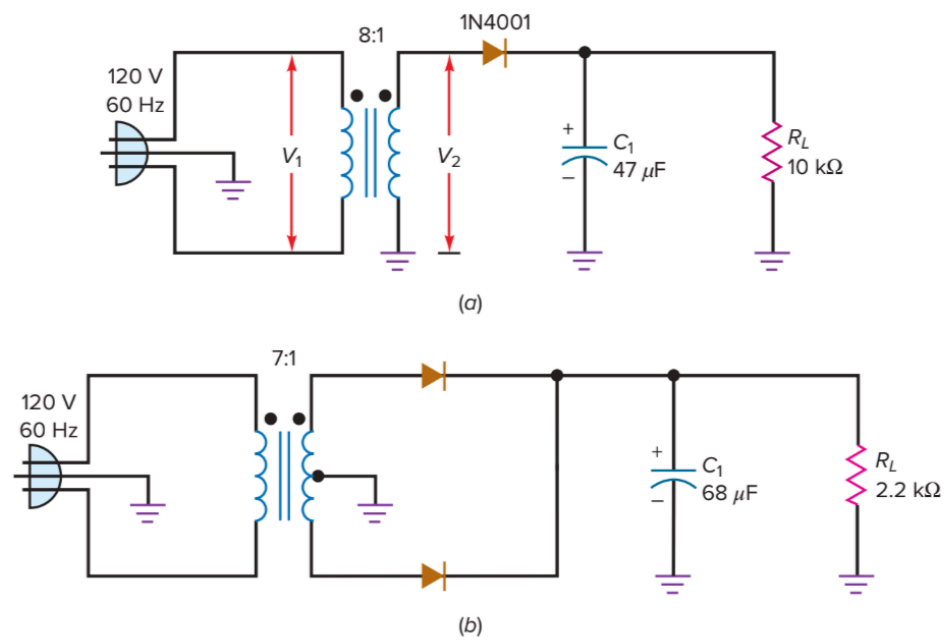


Figure 4-41

